

You must type or neatly handwrite each answer on a separate piece of paper and select the pages for each question when you upload to Gradescope. Feel free to use more than one sheet for an answer if needed. Note that both PDF and picture/photo format (JPG) are accepted. You must show your work. During the quiz, you may consult the textbook, your own notes and other course materials you yourself prepared before the beginning of the quiz. You may not consult resources online. You can post a private question to Ed Discussion to instructors only, but do not make public posts on Ed Discussion, Discord, or other online forums.

By submitting this quiz, you agree to adhere to the statement below: “As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others. All of the work submitted here is mine, and I have not discussed the content of this exam with any other person. I have not consulted any resources beyond the course materials and my notes during the exam. I have not used calculators, computers, or the internet to do calculations or look up answers.”

1. Suppose that A, B , and C are sets such that $B \subseteq C$ and $A \subseteq B$. Prove or disprove that

$$(A \times (A \cup B)) \cap (C \times C) = (A \times A) \cup (A \times B).$$

Solution: We will first show that $(A \times (A \cup B)) \cap (C \times C) \subseteq (A \times A) \cup (A \times B)$.

Suppose that (x, y) is an element of $(A \times (A \cup B)) \cap (C \times C)$. Then we have that $(x, y) \in (A \times (A \cup B))$ and $(x, y) \in (C \times C)$, which gives us that $x \in A$ and $y \in A$ or $y \in B$. It follows that $(x, y) \in A \times A$ or $(x, y) \in A \times B$, and hence $(x, y) \in (A \times A) \cup (A \times B)$ and we have that $(A \times (A \cup B)) \cap (C \times C) \subseteq (A \times A) \cup (A \times B)$.

For the other direction, suppose that $(x, y) \in (A \times A) \cup (A \times B)$. Then $(x, y) \in (A \times A)$ or $(x, y) \in (A \times B)$. If $(x, y) \in A \times A$, then $x \in A$ and $y \in A \subseteq B$ so $(x, y) \in (A \times (A \cup B))$. We also have that $x \in A$ and $y \in A$ where $A \subseteq B \subseteq C$ and so $(x, y) \in C \times C$ as well. Thus we get that $(x, y) \in (A \times (A \cup B)) \cap (C \times C)$. If $(x, y) \in (A \times B)$ then $x \in A$ and $y \in B \subseteq C$ so $(x, y) \in A \times (A \cup B)$ and since $x \in A \subseteq B \subseteq C$ and $y \in B \subseteq C$ we have that x and y are both in C and $(x, y) \in C \times C$. Hence in either case we have that $(x, y) \in (A \times (A \cup B)) \cap (C \times C)$.

It follows that $(A \times (A \cup B)) \cap (C \times C) \subseteq (A \times A) \cup (A \times B)$ and $(A \times (A \cup B)) \cap (C \times C) \supseteq (A \times A) \cup (A \times B)$ which gives the equality $(A \times (A \cup B)) \cap (C \times C) = (A \times A) \cup (A \times B)$.

2. Let $f : A \rightarrow B$ be a function, where A and B are sets. Assume that A contains exactly one element. Show that f is onto if and only if B also contains exactly one element.

Solution: Let a be the element of A . If f is onto, then, for every element $b \in B$, there exists some element $\alpha \in A$ such that $f(\alpha) = b$. But A has only one element, so, for any $b \in B$, we must be able to write $f(a) = b$. Therefore all elements of b are equal to each other, so B contains exactly one element, $f(a)$.

If B contains exactly one element, then that element must be $f(a)$ by construction, so f covers every element of B and is therefore onto.